Citation audit: prior adult guitar-improvisation brief

Scope: Claim-to-source audit of cited factual assertions in the previously delivered research brief; the separate evidence table was checked for duplicate study identities and linked-source metadata. Proposed practice schedules and disproof tests are recommendations, not empirical claims, and were not audited as evidence claims. Full text was accessible for the guitar self-evaluation and clarinet practice-schedule studies; the Hartz & Bauer record remained abstract-level for this check.[1–3]

Summary

Audited **16 atomic claims**.

- **Supported:** 11
- **Partially supported:** 4
- **Citation mismatch:** 1
- **Contradicted:** 0
- **Unverifiable:** 0

Could not verify

No in-scope claim lacked sufficient source access.

Claim-by-claim audit

ID	Section	Exact claim	Cited reference(s)	Access	Verdict	Evidence	Issue	Recommended action
C01	Evidence framing	Direct tests of plateaued intermediate adult guitarists improving expressive improvisation with durable unfamiliar-context transfer were scarce in the retrieved evidence.	[1, 2, 4, 5]	abstract only	Partially supported	The cited records describe broad music-practice coverage, piano leap training, clarinet excerpt practice, and classical-guitar self-evaluation; none is a direct adult intermediate guitar-improvisation transfer trial. The sources illustrate the mismatch but cannot independently prove that no direct study exists.[1, 2, 4, 5]	The evidence-gap statement is based on the search performed, not on a cited source that establishes absence across the literature.	Retain as a scoped search conclusion; avoid treating it as a systematic proof of absence.

ID	Section	Exact claim	Cited reference(s)	Access	Verdict	Evidence	Issue	Recommended action
C02	Multi-causal plateau framing	Improvisation involves musical patterns, audiation, and matching instrument or voice execution to musical intentions.	[3]	abstract only	Partially supported	The source explicitly presents these as conceptual descriptions and quoted expert-pedagogy claims about patterns, audiation, and fluid execution.[3]	The previous prose phrased this as a general empirical fact; the cited material is background/pedagogical framing, not a direct experimental result about plateaued adult guitarists.	Attribute the description to improvisation pedagogy or label it a useful model rather than an established causal finding.
C03	Self-regulated learning evidence	A 2025 systematic review of 31 studies, including seven meta-analyzed, reported a positive medium overall effect for music self-regulated-learning interventions and called for task-aligned outcomes and more informal/professional samples.	[6]	abstract only	Supported	The abstract reports 31 studies, seven included in meta-analysis, a significant positive medium overall effect, and calls for task-aligned outcomes and expanded informal/professional samples.[6]	Abstract-only assessment; no pooled numerical effect estimate was included in the prior brief.	Keep the abstract-level qualifier and do not infer guitar improvisation effects.

ID	Section	Exact claim	Cited reference(s)	Access	Verdict	Evidence	Issue	Recommended action
C04	Music-practice review context	The 2021 PRISMA music-practice review included 296 publications from 3,102 identified records.	[4]	abstract only	Supported	The abstract gives 3,102 records and 296 publications selected for final analysis.[4]	No material citation issue; this review maps music-practice research broadly, not improvisation treatment effects.	Retain with the scope limitation already stated.
C05	Phrase learning versus scales	The Russell study evaluated a cappella harmony singing with tertiary jazz students over three semesters using qualitative observation and interviews, and reported students recognized theory in practice/performance.	[7]	abstract only	Supported	The abstract states the population, three-semester intervention, qualitative methods, and reported recognition of concepts in practice/performance.[7]	No controlled comparison or delayed transfer was reported; the previous brief stated those limits.	Retain as qualitative evidence only.

ID	Section	Exact claim	Cited reference(s)	Access	Verdict	Evidence	Issue	Recommended action
C06	Phrase learning versus scales / duplicated Hartz & Bauer entry	The accessible Hartz & Bauer adult amateur wind-instrument study record does not provide enough sample, intervention, or outcome detail to support an efficacy claim.	[8]	abstract only	Citation mismatch	The cited chunks discuss the broader lack of research on adult musicians and summarize prior ear-playing findings; they do not state this study's sample, intervention, or outcome reporting status.[8]	The exact citation does not substantiate the study-specific access caveat. In addition, xJEAw-BAA and PkSlC6tw are duplicate catalog records for the same Hartz & Bauer (2016) title/authors/URL, not separate studies.	Count Hartz & Bauer once. Describe missing detail as a limitation of the retrieved record, not as a finding from chunks 3 and 7; do not use it as efficacy evidence.

ID	Section	Exact claim	Cited reference(s)	Access	Verdict	Evidence	Issue	Recommended action
C07	Blocked versus variable practice	In the 30-minute pianist leap study, variable practice improved next-day execution time, while immediate distance-error changes were nonsignificant in both groups; transfer was tested on a novel interval but its result was absent from the abstract.	[5]	abstract only	Supported	The abstract states 30-minute training, fixed versus variable interval practice, a novel-interval transfer test, 24-hour retention, nonsignificant immediate distance-error gains, and next-day variable-group execution-time improvement.[5]	The study is a narrow piano motor task and the transfer result is not stated in the indexed abstract.	Retain the task/population and unreported-transfer qualifiers.

ID	Section	Exact claim	Cited reference(s)	Access	Verdict	Evidence	Issue	Recommended action
C08	Blocked versus variable practice	In 10 advanced clarinetists, interleaved excerpts were rated better whenever raters detected a condition difference, including at one day, with variation across raters.	[2]	full text	Supported	The full-text abstract gives N=10, blocked/interleaved excerpt schedules, one-day delayed performance, and the rater-dependent pattern.[2]	No unfamiliar-piece transfer test; the claim is limited to ratings on practiced excerpts.	Retain the comparator and transfer limitation.
C09	Blocked versus variable practice	In a 120-participant university-musician percussion study, varied practice caused more acquisition errors but later retention/transfer accuracy was equivalent to blocked practice.	[9]	abstract only	Supported	The abstract reports N=120, varied-group acquisition errors, equivalent later accuracy, and lower transfer-task scores.[9]	Constrained percussion sticking task and abstract-only access; not improvisation.	Retain the population/task mismatch.

ID	Section	Exact claim	Cited reference(s)	Access	Verdict	Evidence	Issue	Recommended action
C10	Feedback and self-assessment	In a randomized study of 16 college classical guitarists, repeated video review changed self-evaluation comments and strategy-oriented comments, but did not measure improvisation performance or transfer; reported values concern comment categories.	[1]	full text	Supported	The full text specifies the randomized groups, n=16, review schedule, dependent variable as coded comments, reported values, and conclusions on self-evaluation rather than performance transfer.[1]	The measure is behavior/comment categories in written classical repertoire, not improvisation skill.	Keep the explicit distinction between self-evaluation change and performance improvement.

ID	Section	Exact claim	Cited reference(s)	Access	Verdict	Evidence	Issue	Recommended action
C11	Feedback and self-assessment	The 2025 music self-regulation review supports structured planning, strategy teaching, and technology-enhanced third-person feedback as mechanisms, but does not isolate feedback timing/frequency or adult guitar improvisation.	[6]	abstract only	Supported	The abstract describes plan-practice-reflection, strategy teaching, technological third-person feedback, and the need for more task-aligned/further-population evidence.[6]	Abstract-level review findings do not test the specific tutor feedback schedule proposed in the previous brief.	Retain the stated boundary between review mechanisms and guitar-specific efficacy.
C12	Mixed results	The cited piano, clarinet, and percussion studies differ in task and outcomes and do not establish a universal practice-schedule effect for improvisation.	[2, 5, 9]	abstract only	Supported	The sources report distinct tasks and populations, with favorable delayed findings in some measures and null/equivalent retention-transfer findings in another.[2, 5, 9]	No direct improvisation schedule comparison; conclusions are limited to cross-study heterogeneity.	Keep the non-universal and cross-task qualification.

ID	Section	Exact claim	Cited reference(s)	Access	Verdict	Evidence	Issue	Recommended action
C13	Open evidence gap	No reliable head-to-head comparison of isolated ear drills versus integrated listen-sing-play practice in plateaued adult guitarists was located.	[7, 8]	abstract only	Partially supported	R7Jd_PmQ describes qualitative integrated singing work without a control group; xJEAw-BAA concerns adult ear-playing background and earlier studies, not the absence of a head-to-head comparison.[7, 8]	The absence statement reflects the search, and the xJE citation does not substantiate it.	Retain only as a scoped search-gap statement; remove xJEAw-BAA as support for absence.
C14	Challenge 1 counter-argument	A small quasi-experimental study in nine fourth-grade string students found improvisation-based technique learning transferred to an unfamiliar excerpt approximately as well as repertoire-based learning.	[10]	abstract only	Supported	The abstract gives n=9, fourth grade, string students, quasi-experimental comparison, unfamiliar excerpt, and approximately equivalent transfer.[10]	Large population and skill mismatch; the prior brief labels it as a counterargument, not direct adult-guitar evidence.	Retain with the child/string/technique-transfer qualification.

ID	Section	Exact claim	Cited reference(s)	Access	Verdict	Evidence	Issue	Recommended action
C15	Challenge 2 counter-argument	The music self-regulation review supports structured instructional support, but does not establish that learner autonomy is best in every circumstance or test the risk of delayed feedback in guitar improvisation.	[6]	abstract only	Partially supported	The abstract describes structured learning support, strategy instruction, and teacher support; it does not discuss erroneous repetition under delayed feedback.[6]	The source supports a role for structure, but not the specific motor-error counterargument.	Keep only the structured-support component attributed to this source; identify the risk claim as a rationale to test, not a finding.

ID	Section	Exact claim	Cited reference(s)	Access	Verdict	Evidence	Issue	Recommended action
C16	Design conclusion	Immediate practice performance, next-day retention, transfer, and guitarists' self-evaluation behavior can diverge across the cited studies.	[1, 5, 9]	abstract only	Supported	The pianist study separates immediate and next-day leap measures and tested transfer; the percussion study reports acquisition versus later retention/transfer differences; the guitar video-feedback study reports self-evaluation behavior rather than performance.[1, 5, 9]	Cross-study synthesis, not a single within-study demonstration of every outcome diverging.	Retain as a cross-study caution; avoid implying a single experiment compared all four endpoints.

Reference resolution

- **Duplicate**: Hartz & Bauer's 2016 adult amateur wind-instrument study appears under two catalog records with the same title, authors, and URL; count it once.
- The claim-by-claim table records access level and verification status for the prior brief's other cited studies.

Limitations

- Claims assessed from abstract only: 14
- Unresolved or ambiguous bibliography entries: 0
- Claims marked unverifiable: 0

References

1. Boucher M, Creech A, Dubé F (2019) Video feedback and the self-evaluation of college-level guitarists during individual practice. Psychology of Music. https://doi.org/10.1177/0305735619842374

2. Carter C, Grahn JA (2016) Optimizing Music Learning: Exploring How Blocked and Interleaved Practice Schedules Affect Advanced Performance. Frontiers in Psychology. https://doi.org/10.3389/fpsyg.2016.01251

3. Hartz B, Bauer WR (2016) The Effect of Ear Playing Instruction on Adult Amateur Wind Instrumentalists' Musical Self-Efficacy: An Exploratory Study. Contributions to Music Education

4. How ER, Tan L, Miksza P (2021) A PRISMA review of research on music practice. Music & Science. https://doi.org/10.1177/10298649211005531

5. Bangert M, Wiedemann A, Jabusch H, Weber C (2013) When less of the same is more: Benefits of variability of practice in pianists

6. Wang Y, Zhang M, Zhang H, et al (2025) From Practice to Reflection: A Systematic Review of Mechanisms Driving Metacognition and SRL in Music. Journal of Intelligence. https://doi.org/10.3390/jintelligence13120162

7. Russell H (2017) A cappella ear training: Bringing theory and aural skills together via singing in a jazz program environment. Australian journal of music education

8. Hartz B, Bauer WR (2016) The Effect of Ear Playing Instruction on Adult Amateur Wind Instrumentalists' Musical Self-Efficacy: An Exploratory Study. Contributions to Music Education

9. Rose L (2006) The effects of contextual interference on the acquisition, retention, and transfer of a music motor skill among university musicians. https://doi.org/10.31390/gradschool_dissertations.1643

10. Minnich SW (2014) THE EFFECT OF IMPROVISATION ON THE TRANSFER OF TECHNIQUE - Sally Williams Minnich